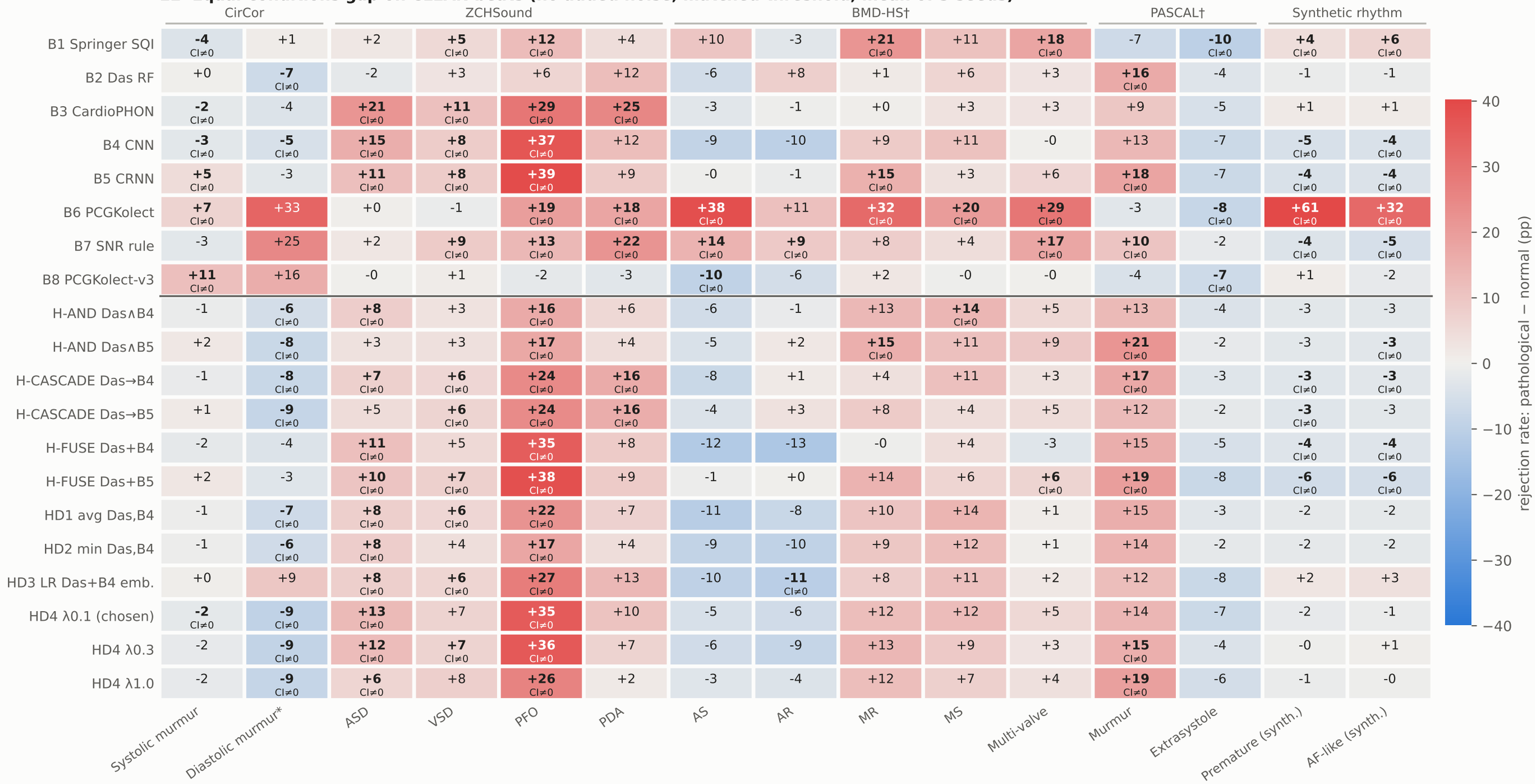
E2 Equal-conditions gap on CLEAN beats (no added noise, matched threshold, mean of 3 seeds)



	Systolic murmur	Diastolic murmur*	ASD	VSD	PFO	PDA	AS	AR	MR	MS	Multi-valve	Murmur	Extrasystole	Premature (synth.)	AF-like (synth.)
B1 Springer SQI	**-4** CI≠0	+1	+2	**+5** CI≠0	**+12** CI≠0	+4	+10	-3	**+21** CI≠0	+11	**+18** CI≠0	-7	**-10** CI≠0	**+4** CI≠0	**+6** CI≠0
B2 Das RF	+0	**-7** CI≠0	-2	+3	+6	+12	-6	+8	+1	+6	+3	**+16** CI≠0	-4	-1	-1
B3 CardioPHON	**-2** CI≠0	-4	**+21** CI≠0	**+11** CI≠0	**+29** CI≠0	**+25** CI≠0	-3	-1	+0	+3	+3	+9	-5	+1	+1
B4 CNN	**-3** CI≠0	**-5** CI≠0	**+15** CI≠0	**+8** CI≠0	**+37** CI≠0	+12	-9	-10	+9	+11	-0	+13	-7	**-5** CI≠0	**-4** CI≠0
B5 CRNN	**+5** CI≠0	-3	**+11** CI≠0	**+8** CI≠0	**+39** CI≠0	+9	-0	-1	**+15** CI≠0	+3	+6	**+18** CI≠0	-7	**-4** CI≠0	**-4** CI≠0
B6 PCGKolect	**+7** CI≠0	+33	+0	-1	**+19** CI≠0	**+18** CI≠0	**+38** CI≠0	+11	**+32** CI≠0	**+20** CI≠0	**+29** CI≠0	-3	**-8** CI≠0	**+61** CI≠0	**+32** CI≠0
B7 SNR rule	-3	+25	+2	**+9** CI≠0	**+13** CI≠0	**+22** CI≠0	**+14** CI≠0	**+9** CI≠0	+8	+4	**+17** CI≠0	**+10** CI≠0	-2	**-4** CI≠0	**-5** CI≠0
B8 PCGKolect-v3	**+11** CI≠0	+16	-0	+1	-2	-3	**-10** CI≠0	-6	+2	-0	-0	-4	**-7** CI≠0	+1	-2
H-AND Das∧B4	-1	**-6** CI≠0	**+8** CI≠0	+3	**+16** CI≠0	+6	-6	-1	+13	**+14** CI≠0	+5	+13	-4	-3	-3
H-AND Das∧B5	+2	**-8** CI≠0	+3	+3	**+17** CI≠0	+4	-5	+2	**+15** CI≠0	+11	+9	**+21** CI≠0	-2	-3	**-3** CI≠0
H-CASCADE Das→B4	-1	**-8** CI≠0	**+7** CI≠0	**+6** CI≠0	**+24** CI≠0	**+16** CI≠0	-8	+1	+4	+11	+3	**+17** CI≠0	-3	**-3** CI≠0	**-3** CI≠0
H-CASCADE Das→B5	+1	**-9** CI≠0	+5	**+6** CI≠0	**+24** CI≠0	**+16** CI≠0	-4	+3	+8	+4	+5	+12	-2	**-3** CI≠0	-3
H-FUSE Das+B4	-2	-4	**+11** CI≠0	+5	**+35** CI≠0	+8	-12	-13	-0	+4	-3	+15	-5	**-4** CI≠0	**-4** CI≠0
H-FUSE Das+B5	+2	-3	**+10** CI≠0	**+7** CI≠0	**+38** CI≠0	+9	-1	+0	+14	+6	**+6** CI≠0	**+19** CI≠0	-8	**-6** CI≠0	**-6** CI≠0
HD1 avg Das,B4	-1	**-7** CI≠0	**+8** CI≠0	**+6** CI≠0	**+22** CI≠0	+7	-11	-8	+10	+14	+1	+15	-3	-2	-2
HD2 min Das,B4	-1	**-6** CI≠0	**+8** CI≠0	+4	**+17** CI≠0	+4	-9	-10	+9	+12	+1	+14	-2	-2	-2
HD3 LR Das+B4 emb.	+0	+9	**+8** CI≠0	**+6** CI≠0	**+27** CI≠0	+13	-10	**-11** CI≠0	+8	+11	+2	+12	-8	+2	+3
HD4 λ0.1 (chosen)	**-2** CI≠0	**-9** CI≠0	**+13** CI≠0	+7	**+35** CI≠0	+10	-5	-6	+12	+12	+5	+14	-7	-2	-1
HD4 λ0.3	-2	**-9** CI≠0	**+12** CI≠0	**+7** CI≠0	**+36** CI≠0	+7	-6	-9	+13	+9	+3	**+15** CI≠0	-4	-0	+1
HD4 λ1.0	-2	**-9** CI≠0	**+6** CI≠0	+8	**+26** CI≠0	+2	-3	-4	+12	+7	+4	**+19** CI≠0	-6	-1	-0

Same beats quality-matched by the clean rule. Bold/"CI≠0" = 95 % patient-bootstrap CI excludes 0 in ≥ 2 of 3 seeds. Rows below the line: hybrids.